HOCHSCHULE RAVENSBURG-WEINGARTEN UNIVERSITY OF APPLIED SCIENCES

GPIO Output Stages in Simulation

Push-pull, open drain and open collector: switching speed, voltage and current limits

Embedded Computing – Master of Engineering – Worksheet with solutions

Overview

Every GPIO pin ends in an output stage: two transistors (push-pull) or one transistor to ground (open drain, open collector). In this worksheet you will take these stages apart in a circuit simulator. You will find out how much current a pin can deliver, how fast it can switch a load, why a pull-up resistor makes a bus slow, and which voltages and currents destroy a pin or the circuit behind it.

Learning outcomes. After this worksheet you can

- model a GPIO output stage by on-resistance and saturation current and relate the model to the V_{OH}/V_{OL} specifications of a data sheet,
- predict rise and fall times, the maximum toggle frequency and the dynamic power of a push-pull output,
- dimension the pull-up resistor of an open-drain bus such as I^2C from rise-time and V_{OL} limits,
- judge where the maximum output voltage and current of a pin come from (body diodes, injection current, back-powering, transistor breakdown and saturation) and protect a driver against inductive kick-back,
- recognise when a PCB trace has to be treated as a transmission line and terminate it.

Organisation. About 4 hours, in teams of two. You do not have to calculate: all results of the model are given in tables. Tasks marked [P] (preparation) ask you to read these values and understand where they come from *before* you open the simulation. [S] means simulate, measure and enter your values in the empty column *Simulation* of the tables. [D] means discuss and explain. Hand in a short report (PDF): for each task the completed tables, screenshots of the scopes and your explanation. When a measurement differs from the model value by more than about $10\,\%$, explain why.

Tool. Falstad CircuitJS runs in the browser: `https://www.falstad.com/circuit/`. Each part of this sheet has a link that opens the prepared circuit. The circuit files are also in the folder `examples/interfaces/falstad/` of the lecture repository (open them with *File* → *Open File*).

Working with CircuitJS

- Hover over an element to see its voltages, currents and power in the lower right corner.
- Click on a logic input (H/L) or a switch to toggle it. Double-click an element to edit its value. Sliders on the right change the values marked in the circuit.
- Scopes: right-click → *Properties* sets the time base ("scope speed"), the scale and shown values (e.g. average, peak-to-peak). Drag with the mouse in a scope to measure time differences.
- *Reset* restarts the simulation, *Run/STOP* freezes it for reading values.
- *Options* → *Other Options* lets you change the simulation time step. If edges look jagged, make the time step smaller.

The transistor model used in all circuits

The MOSFETs follow the square-law (Shichman–Hodges) model with threshold $V_T = 0.6\,\text{V}$ and

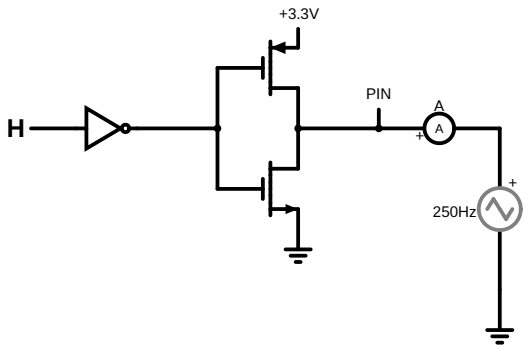
$$I_D = \begin{cases} \beta\left[(V_{GS} - V_T)\,V_{DS} - \frac{1}{2}V_{DS}^2\right] & V_{DS} < V_{GS} - V_T \quad \text{(triode)} \\ \frac{\beta}{2}\,(V_{GS} - V_T)^2 & V_{DS} \geq V_{GS} - V_T \quad \text{(saturation)} \end{cases}$$

with $\beta_N = 12\,\text{mA/V}^2$ for the NMOS and $\beta_P = 8\,\text{mA/V}^2$ for the PMOS of the output stage (PMOS: use V_{SG}, V_{SD}). Both have a body diode. The supply is $V_{DD} = 3.3\,\text{V}$. The model knows no gate or junction capacitances, no temperature and no breakdown. Pin and load capacitances are drawn as explicit capacitors. Keep in mind what the model cannot show when you interpret the results.

1 Push-pull output: static characteristic

▶ open in Falstad (`A_pushpull_characteristic.txt`)

The logic input is the bit in the output register. Two inverters (the predriver and the CMOS stage) make the pin follow it. A triangle voltage source acts as a curve tracer: it forces the pin voltage from $-0.5\,\text{V}$ to $3.8\,\text{V}$, and an ammeter measures the pin current (positive = current out of the pin). The left scope shows the current over the voltage, the right scope the power in both transistors.



Task 1.1 [P] Table 1.1 gives the on-resistance (valid for small V_{DS}) and the saturation current of both transistors for $V_{GS} = V_{DD}$. Which one is the stronger transistor, and why do chip designers make the PMOS larger than the NMOS?

Table 1.1: model values of the output transistors ($V_{GS} = 3.3\,\text{V}$)

	formula	NMOS	PMOS
β		$12\,\text{mA/V}^2$	$8\,\text{mA/V}^2$
R_{on}	$1/(\beta(V_{GS} - V_T))$	$30.9\,\Omega$	$46.3\,\Omega$
I_{sat}	$\frac{\beta}{2}(V_{GS} - V_T)^2$	$43.7\,\text{mA}$	$29.2\,\text{mA}$

Solution

Step 1: Gate overdrive: with $V_{GS} = V_{DD} = 3.3\,\text{V}$ and $V_T = 0.6\,\text{V}$ both transistors have $V_{GS} - V_T = 2.7\,\text{V}$.

Step 2: On-resistance: for small V_{DS} the triode equation reduces to $I_D \approx \beta(V_{GS} - V_T)V_{DS}$, so $R_{on} =$

$V_{DS}/I_D = 1/(\beta(V_{GS} - V_T))$:
$R_{on,N} = 1/(12\,\text{mA/V}^2 \cdot 2.7\,\text{V}) = 30.9\,\Omega$, $R_{on,P} = 1/(8\,\text{mA/V}^2 \cdot 2.7\,\text{V}) = 46.3\,\Omega$.

Step 3: Saturation current: $I_{sat} = \frac{\beta}{2}(V_{GS} - V_T)^2$ with $(2.7\,\text{V})^2 = 7.29\,\text{V}^2$:
$I_{sat,N} = 6\,\text{mA/V}^2 \cdot 7.29\,\text{V}^2 = 43.7\,\text{mA}$, $I_{sat,P} = 4\,\text{mA/V}^2 \cdot 7.29\,\text{V}^2 = 29.2\,\text{mA}$.

Step 4: Comparison: the NMOS has the lower R_{on} and the higher I_{sat}, both by the factor $\beta_N/\beta_P = 1.5$.

Step 5: Reason: $\beta = \mu C_{ox} W/L$. The mobility of holes (PMOS) is about 2–3 times lower than that of electrons (NMOS), so a PMOS needs a 2–3 times wider channel W for the same β. Here the PMOS is larger, but not enough to fully compensate, which saves chip area.

Answer: The NMOS is the stronger transistor. Designers make the PMOS wider to compensate for the lower hole mobility. It often stays somewhat weaker, which is why data sheets allow a larger V_{OH} drop than V_{OL} rise, or specify asymmetric currents.

Task 1.2 [S] Set the output to H and then to L. From the curve, read V_{OH} and V_{OL} at the currents of Table 1.2 (Run/STOP and hovering over the PIN label helps) and enter them next to the model values. Also read the short-circuit currents (pin at 0 V for H, pin at 3.3 V for L).

Table 1.2: output levels over the load current

$\lvert I\rvert$	2 mA	4 mA	8 mA	12 mA	20 mA	short circuit
V_{OH}, model	3.21 V	3.11 V	2.90 V	2.67 V	2.11 V	29.2 mA
V_{OH}, simulation	3.21	3.11	2.90	2.67	2.11	29.2
V_{OL}, model	0.06 V	0.13 V	0.26 V	0.40 V	0.71 V	43.7 mA
V_{OL}, simulation	0.06	0.13	0.26	0.40	0.71	43.7

Solution

Step 1: Output H: the PMOS conducts, the pin current flows out of the pin and the voltage across the PMOS is $V_{SD} = V_{DD} - V_{OH}$. For small currents the PMOS is in the triode region: $I = \beta_P\left[2.7\,V_{SD} - \frac{1}{2}V_{SD}^2\right]$.

Step 2: Solve the quadratic equation for V_{SD}: $V_{SD} = 2.7\,\text{V} - \sqrt{(2.7\,\text{V})^2 - 2I/\beta}$. Example $I = 8$ mA: $2I/\beta_P = 2\,\text{V}^2$, $V_{SD} = 2.7 - \sqrt{5.29} = 0.40$ V, so $V_{OH} = 2.90$ V.

Step 3: Output L: the same formula with β_N gives $V_{OL} = V_{DS}$. Example $I = 8$ mA: $2I/\beta_N = 1.33\,\text{V}^2$, $V_{OL} = 2.7 - \sqrt{5.96} = 0.26$ V.

Step 4: In the simulation: set the logic input, press Run/STOP while the sweep passes the desired current and hover over PIN. The measured values are:

$\lvert I\rvert$ / mA	2	4	8	12	20
V_{OH} / V	3.21	3.11	2.90	2.67	2.11
V_{OL} / V	0.06	0.13	0.26	0.40	0.71

Step 5: Short circuit: for H with the pin at 0 V the PMOS sees $V_{SD} = 3.3\,\text{V} > V_{GS} - V_T$, so it is saturated and delivers $I_{sat,P} = 29.2$ mA. For L with the pin at 3.3 V the NMOS delivers $I_{sat,N} = 43.7$ mA.

Answer: Model and simulation agree to within the reading accuracy. Short-circuit currents: 29.2 mA (high, PMOS saturated) and 43.7 mA (low, NMOS saturated).

Task 1.3 [D] The RP2040 data sheet specifies $V_{OH} \geq 2.62\,\text{V}$ and $V_{OL} \leq 0.5\,\text{V}$ at a load current equal to the selected drive strength (2, 4, 8 or 12 mA) and $IOVDD = 3.3\,\text{V}$. The model delivers $12.8\,\text{mA}$ at $V_{OH} = 2.62\,\text{V}$ and sinks $12\,\text{mA}$ at $V_{OL} = 0.40\,\text{V}$. Which drive strength setting does the model represent? Drive strength is implemented by switching transistor "fingers" in parallel, i.e. by scaling β ($I_D \propto \beta$): the 4 mA setting needs $\beta_P = 2.5\,\text{mA/V}^2$. What fraction of the fingers is that, and what does it mean for the switching speed (part 3)?

Solution

Step 1: High side: at $V_{OH} = 2.62\,\text{V}$ the PMOS has $V_{SD} = 0.68\,\text{V}$ and delivers $I = 8\,\text{mA/V}^2 \cdot (2.7 \cdot 0.68 - \frac{1}{2} \cdot 0.68^2)\,\text{V}^2 = 12.8\,\text{mA} \geq 12\,\text{mA}$.

Step 2: Low side: at $V_{OL} = 0.4\,\text{V}$ the NMOS sinks $I = 12\,\text{mA/V}^2 \cdot (2.7 \cdot 0.4 - \frac{1}{2} \cdot 0.4^2)\,\text{V}^2 = 12\,\text{mA}$, exactly the required value.

Step 3: Both data-sheet conditions are met at 12 mA (and not at a higher current), so the model represents the 12 mA setting.

Step 4: 4 mA setting: at the fixed operating point $V_{SD} = 0.68\,\text{V}$ the current scales with β: $\beta_P = 8\,\text{mA/V}^2 \cdot 4\,\text{mA}/12.8\,\text{mA} = 2.5\,\text{mA/V}^2$. That is $2.5/8 \approx 1/3$ of the fingers.

Step 5: Switching speed: with one third of β, R_{on} is three times larger and I_{sat} three times smaller. Edge times are $\propto C_L/\beta$ (part 3), so the pin switches about three times more slowly.

Answer: The model represents the 12 mA setting. The 4 mA setting uses about one third of the fingers and makes rise and fall times about three times longer for the same load capacitance.

Task 1.4 [S/D] A red LED ($V_F \approx 2.0\,\text{V}$) with a $100\,\Omega$ series resistor is connected from the pin to GND. The naive calculation gives $I = (3.3\,\text{V} - 2.0\,\text{V})/100\,\Omega = 13\,\text{mA}$. Draw the load line $I = (V_{PIN} - V_F)/R$ into your measured characteristic (it passes through 2.0 V/0 mA and 3.3 V/13 mA) and read the operating point. The model gives $V_{PIN} = 2.86\,\text{V}$ and $I = 8.6\,\text{mA}$. Why is the current a third lower than the naive value? Optional: verify it by replacing the sweep source with an LED and a resistor (*Draw → Outputs and Labels → Add LED*).

Solution

Step 1: Load line: the LED branch draws $I = (V_{PIN} - 2.0\,\text{V})/100\,\Omega$. Draw the straight line through 2.0 V/0 mA and 3.3 V/13 mA into the H curve.

Step 2: Operating point: the pin current must satisfy both the output characteristic and the load line. With $V_{SD} = V_{DD} - V_{PIN}$: $10\,\text{mA/V} \cdot (1.3\,\text{V} - V_{SD}) = 8\,\text{mA/V}^2\,(2.7\,V_{SD} - \frac{1}{2}V_{SD}^2) \Rightarrow 4V_{SD}^2 - 31.6\,V_{SD} + 13 = 0 \Rightarrow V_{SD} = 0.44\,\text{V}$.

Step 3: Result: $V_{PIN} = 2.86\,\text{V}$, $I = (2.86 - 2.0)\,\text{V}/100\,\Omega = 8.6\,\text{mA}$. The graphical intersection in the simulation gives the same values.

Step 4: Interpretation: at this point the PMOS behaves like a resistor of $0.44\,\text{V}/8.6\,\text{mA} \approx 50\,\Omega$, which is in series with the $100\,\Omega$. The total resistance is $150\,\Omega$ instead of $100\,\Omega$, so the current drops to about two thirds.

Answer: The output resistance of the pin (about $50\,\Omega$) is not negligible compared to $100\,\Omega$. For a defined LED current either choose a larger series resistor (then the pin resistance becomes negligible) or include the output stage in the calculation.

Task 1.5 [D] Look at the curve below 0 V and above 3.3 V. Which element conducts there? Relate this to the absolute maximum rating "pin voltage $-0.5 \ldots IOVDD + 0.5\,\text{V}$" of the RP2040 and to the term *injection current*. Why can a large injection current upset an analogue measurement or even a digital circuit on the same chip?

Solution

Step 1: Above V_{DD}: the drain of the PMOS (the pin) becomes more positive than its source and body (V_{DD}). The body diode drain→body is forward biased and conducts current from the pin into the supply. In a real chip the ESD clamp diode to V_{DD} is in parallel.

Step 2: Below 0 V: the body diode of the NMOS (body at GND, drain at the pin) is forward biased and conducts current from GND out of the pin.

Step 3: Shape of the curve: from about 0.4 V beyond the rails the diode current rises exponentially (in the model to about 9 mA at 3.7 V).

Step 4: Absolute maximum rating: the limits $-0.5 \ldots IOVDD + 0.5$ V keep these diodes almost off. The current that still flows through them is called injection current.

Step 5: Consequences: injected current flows as minority carriers into the substrate or into the supply rail. It can shift the ADC reference and neighbouring analogue channels, trigger the parasitic thyristor of the CMOS structure (latch-up, a short circuit between V_{DD} and GND), or raise V_{DD} (see part 5).

Answer: The body diodes (and ESD diodes) conduct outside the supply range. The absolute maximum rating keeps them off; a large injection current disturbs analogue measurements and can latch up the chip.

Task 1.6 [S/D] Read the maximum power in the transistors from the right scope (model: 96 mW in the PMOS for a short circuit to GND, 110 mW at -0.5 V). Many microcontrollers allow 50 mA to 100 mA in total for all pins of one supply domain. What does this mean for the number of pins that may drive LEDs at the same time, and what happens to the chip temperature if a pin is shorted to GND?

Solution

Step 1: Short circuit to GND: the PMOS is saturated, carries 29.2 mA and has the full supply across it: $P = 3.3\,\text{V} \cdot 29.2\,\text{mA} \approx 96\,\text{mW}$.

Step 2: Pin at -0.5 V: the current stays at I_{sat}, but the voltage rises to 3.8 V: $P \approx 3.8\,\text{V} \cdot 29.2\,\text{mA} \approx 110\,\text{mW}$.

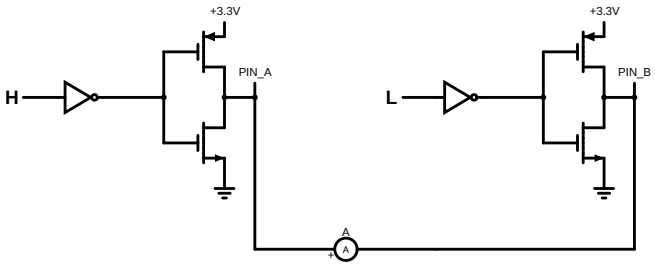
Step 3: Number of LEDs: each LED from task 1.4 draws 8.6 mA. With a budget of 50 mA per supply domain: $50\,\text{mA}/8.6\,\text{mA} = 5.8$, so at most five LEDs at the same time (with 100 mA: eleven).

Step 4: Temperature: the 96 mW to 110 mW are dissipated in a tiny transistor area at the pad. The chip as a whole heats up only a little, but locally the temperature rises strongly. The current also flows through thin supply lines on the chip (electromigration).

Answer: Only a few pins may drive LEDs at the same time (about five at 50 mA). A single short-circuited pin usually survives, but it heats up locally and ages. This is why per-pin and per-domain limits exist, and larger loads need an external driver (part 6).

2 Two push-pull outputs on one line

▶ **open in Falstad** (`C_bus_contention.txt`)



Task 2.1 [P/S] Output A drives high, output B drives low. The line voltage follows from $I_{D,P,A}(V) = I_{D,N,B}(V)$; both transistors work in the triode region. Measure the values of Table 2.1 in the simulation.

Table 2.1: contention between two push-pull outputs

	line voltage	current	power PMOS A	power NMOS B
model	$1.09\,\text{V}$	$28.2\,\text{mA}$	$62\,\text{mW}$	$31\,\text{mW}$
simulation	1.09	28.2	62.3	30.8

Solution

Step 1: Current path: from V_{DD} through the PMOS of A, along the line, through the NMOS of B to GND. Both transistors carry the same current I.

Step 2: Equation: the PMOS of A sees $V_{SD} = 3.3\,\text{V} - V$, the NMOS of B sees $V_{DS} = V$. Assuming the triode region for both: $\beta_P \left[2.7\,(3.3 - V) - \frac{1}{2}(3.3 - V)^2\right] = \beta_N \left[2.7\,V - \frac{1}{2}V^2\right]$.

Step 3: Solving (numerically, or by trying values) gives $V = 1.09\,\text{V}$. Check: NMOS $12\,\text{mA/V}^2 \cdot (2.7 \cdot 1.09 - \frac{1}{2} \cdot 1.09^2)\,\text{V}^2 = 28.2\,\text{mA}$; PMOS with $V_{SD} = 2.21\,\text{V}$: $8\,\text{mA/V}^2 \cdot (2.7 \cdot 2.21 - \frac{1}{2} \cdot 2.21^2)\,\text{V}^2 = 28.2\,\text{mA}$.

Step 4: Check the assumption: $V_{DS} = 1.09\,\text{V}$ and $V_{SD} = 2.21\,\text{V}$ are both below $V_{GS} - V_T = 2.7\,\text{V}$, so both transistors are indeed in the triode region.

Step 5: Power: $P_{P,A} = 2.21\,\text{V} \cdot 28.2\,\text{mA} = 62\,\text{mW}$, $P_{N,B} = 1.09\,\text{V} \cdot 28.2\,\text{mA} = 31\,\text{mW}$. The weaker PMOS takes the larger voltage and twice the power.

Step 6: Simulation: $1.09\,\text{V}$, $28.20\,\text{mA}$, $62.3\,\text{mW}$ and $30.8\,\text{mW}$.

Answer: The line settles at $1.09\,\text{V}$ with $28.2\,\text{mA}$ contention current; model and simulation agree.

Task 2.2 [D] What logic level does a receiver see on the line?

Solution

Step 1: Input thresholds of a $3.3\,\text{V}$ CMOS input: $V_{IL,max} \approx 0.8\,\text{V}$ (LVTTL) or $0.3\,V_{DD} = 1.0\,\text{V}$, $V_{IH,min} \approx 2.0\,\text{V}$ or $0.7\,V_{DD} = 2.3\,\text{V}$.

Step 2: The line voltage $1.09\,\text{V}$ is above $V_{IL,max}$ and below $V_{IH,min}$.

Step 3: In this range the receiver may read 0 or 1, the value can differ between receivers and change with

temperature, and the input stage draws cross current (task 3.5).

Answer: The level is undefined: $1.09\,\mathrm{V}$ lies in the forbidden zone between $V_{IL,max}$ and $V_{IH,min}$.

Task 2.3 [D] Name three situations in embedded practice where this happens. How do tri-state outputs, open-drain outputs and series resistors prevent the damage? (A series resistor of about $290\,\Omega$ in each output limits the contention current to $5\,\mathrm{mA}$: $2R + R_{on,P} + R_{on,N} = 3.3\,\mathrm{V}/5\,\mathrm{mA}$. What does it cost?)

Solution

Step 1: Situations (any three): two masters on a shared line (bus arbitration error); a bidirectional data line where both sides drive at the same time during the turn-around; bit-banged $\mathrm{I}^2\mathrm{C}$ with the pin configured push-pull; a debugger and the board both driving reset; two boards connected TX to TX; a pin shorted to its neighbour by a solder bridge; a pin configured as output by mistake while an external signal drives it.

Step 2: Tri-state outputs: each driver can switch to high impedance. Only one driver is enabled at a time; the protocol (or the firmware) must guarantee this, so a bug still causes contention.

Step 3: Open-drain outputs: no output drives high, only the pull-up does. If two outputs disagree, the low output wins and the current is limited by the pull-up resistor. Contention is harmless by design.

Step 4: Series resistors: the contention current flows through $2R + R_{on,P} + R_{on,N}$. For $5\,\mathrm{mA}$: $2R + 46.3\,\Omega + 30.9\,\Omega = 3.3\,\mathrm{V}/5\,\mathrm{mA} = 660\,\Omega \Rightarrow R \approx 290\,\Omega$ per output (standard value $330\,\Omega$).

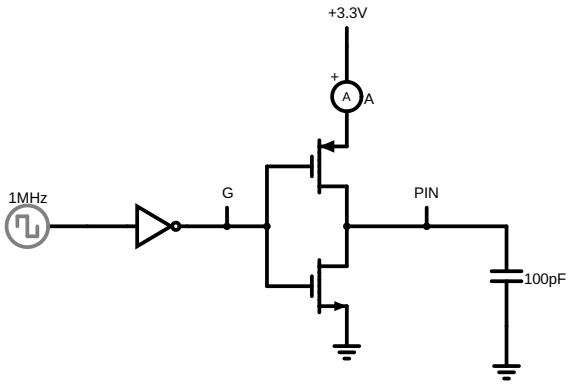
Step 5: Cost: $330\,\Omega$ in series with the load capacitance increases the time constant about tenfold (compared to $R_{on} \approx 30\,\Omega \ldots 46\,\Omega$), and the resistor drops voltage under DC load.

Answer: Contention happens whenever two drivers may be active on one line. Tri-state prevents it by protocol, open drain by design, series resistors by limiting the current, at the cost of slower edges (part 3).

3 Push-pull output: switching speed and dynamic power

▶ open in Falstad (`B_pushpull_capacitive_load.txt`)

The square wave is the program toggling the pin. The predriver is a logic inverter with an adjustable slew rate (how fast the gate voltage changes). The output drives the load capacitance C_L (pin, trace, inputs of other chips). An ammeter measures the current taken from the supply.



Task 3.1 [P] Table 3.1 gives the rise and fall times (10 % to 90 %) from two models: (a) the RC approximation $t_r = 2.2\,R_{on}C_L$, (b) the square-law model, in which the transistor starts in saturation (constant current I_{sat}) and enters the triode region when $V_{DS} = V_{GS} - V_T$ (integration of $dt = C_L\,dV/I_D(V)$). Why is the RC approximation too optimistic?

Solution

Step 1: RC model (a): charging through a constant resistance gives $V(t) = V_{DD}(1 - e^{-t/\tau})$; from 10 % to 90 % it takes $\tau \ln 9 = 2.2\,\tau$ with $\tau = R_{on}C_L$. Rising edge (PMOS): $t_r = 2.2 \cdot 46.3\,\Omega \cdot C_L$, falling edge (NMOS): $t_f = 2.2 \cdot 30.9\,\Omega \cdot C_L$. For 10 pF, 100 pF, 1 nF: 1.0 ns/0.68 ns, 10.2 ns/6.8 ns, 102 ns/68 ns.

Step 2: Square-law model (b), rising edge: the pin starts at 0 V, so the PMOS starts with $V_{SD} = 3.3\,\text{V} > a = V_{GS} - V_T = 2.7\,\text{V}$, i.e. in saturation. It stays saturated until $V_{SD} = 2.7\,\text{V}$, i.e. until the pin reaches 0.6 V.

Step 3: Saturation phase: constant current, so the pin voltage rises linearly. From the 10 % point (0.33 V) to 0.6 V: $t_1 = C_L \cdot 0.27\,\text{V}/29.2\,\text{mA}$ (0.92 ns for 100 pF).

Step 4: Triode phase: $C_L\,dV = I_D\,dt$ with $x = V_{SD}$ falling from 2.7 V to 0.33 V (90 % point):

$$t_2 = \frac{C_L}{\beta}\int_{0.33}^{2.7}\frac{dx}{x(a-x/2)} = \frac{C_L}{\beta a}\left[\ln\frac{x}{2a-x}\right]_{0.33}^{2.7} = \frac{C_L}{\beta}\cdot 1.012\,\text{V}^{-1}.$$

For 100 pF: $t_2 = 100\,\text{pF}/8\,\text{mA/V}^2 \cdot 1.012\,\text{V}^{-1} = 12.65$ ns.

Step 5: Rise time: $t_r = t_1 + t_2 = 13.6$ ns for 100 pF; it scales with C_L: 1.36 ns, 13.6 ns, 136 ns.

Step 6: Falling edge: the same calculation with the NMOS ($I_{sat} = 43.7$ mA, $\beta = 12\,\text{mA/V}^2$): $t_1 = 0.62$ ns, $t_2 = 8.43$ ns for 100 pF, so $t_f = 0.90$ ns, 9.0 ns, 90 ns.

Step 7: Comparison: model (b) is about 33 % slower than model (a), i.e. the RC estimate is about 25 % too short.

Answer: The RC model assumes the transistor is a resistor from the very beginning, so at the start of the edge it would deliver $V_{DD}/R_{on} = 71$ mA (PMOS). In reality the transistor is saturated and delivers only $I_{sat} = 29.2$ mA until V_{DS} has dropped to $V_{GS} - V_T$.

Task 3.2 [S] Set the predriver slew rate to its maximum and measure t_r and t_f for the three capacitances (set the scope time base accordingly, or zoom in with Run/STOP). For which C_L does the predriver dominate the edge?

Table 3.1: rise and fall times at the pin

	$C_L = 10$ pF		$C_L = 100$ pF		$C_L = 1$ nF	
	t_r	t_f	t_r	t_f	t_r	t_f
(a) RC model	1.0 ns	0.68 ns	10.2 ns	6.8 ns	102 ns	68 ns
(b) square-law model	1.36 ns	0.90 ns	13.6 ns	9.0 ns	136 ns	90 ns
simulation	1.5	1.1	13.6	9.1	136	90

Solution

Step 1: Set the slew rate slider to its maximum (2 V/ns or more) and the scope time base to a few edge times. Measure from 0.33 V to 2.97 V (10 % to 90 %).

Step 2: Measured values: 10 pF: 1.5 ns/1.1 ns; 100 pF: 13.6 ns/9.1 ns; 1 nF: 136 ns/90 ns.

Step 3: Comparison: for 100 pF and 1 nF the simulation matches model (b) exactly. For 10 pF it is about $10\,\%$ (rising) and $20\,\%$ (falling) slower than model (b).

Step 4: Reason: the gate voltage itself needs time to swing (3.3 V at 2 V/ns: about 1.5 ns). Model (b) assumes an instantaneous gate step. As long as the output edge is much longer than the gate edge, this does not matter.

Answer: At 10 pF (and below) the predriver dominates the edge; from about 100 pF on the output transistors and C_L determine it.

Task 3.3 [S/D] Increase the toggle frequency with $C_L = 1$ nF until the pin no longer reaches valid logic levels ($V_{IH} = 0.7\,V_{DD}$, $V_{IL} = 0.3\,V_{DD}$). Compare with the rule of thumb $f_{max} \approx 1/(2\,t_r) \approx 3.7$ MHz (slower edge). Why do data sheets specify the maximum GPIO frequency only together with a load capacitance?

Solution

Step 1: Thresholds: $V_{IH} = 0.7 \cdot 3.3\,\text{V} = 2.31\,\text{V}$, $V_{IL} = 0.3 \cdot 3.3\,\text{V} = 0.99\,\text{V}$. The pin must exceed 2.31 V in every high phase and fall below 0.99 V in every low phase.

Step 2: Rule of thumb: each half period must be at least as long as the slower edge, here $t_r = 136$ ns: $f_{max} \approx 1/(2 \cdot 136\,\text{ns}) = 3.7$ MHz.

Step 3: Simulation: up to about 3.7 MHz the pin still reaches almost the full swing. Above that the edges turn into exponential segments and the swing shrinks around $V_{DD}/2$. At about 5 MHz to 6 MHz the levels no longer reach V_{IH} and V_{IL}.

Step 4: The rule of thumb is slightly pessimistic, because valid levels need only $70\,\%$ and $30\,\%$ of the swing, not $90\,\%$ and $10\,\%$.

Answer: The edge times are proportional to C_L, so the maximum frequency is inversely proportional to C_L. A frequency without a load capacitance is meaningless; data sheets therefore state e.g. "50 MHz at 30 pF".

Task 3.4 [S] Set $f = 1$ MHz and the fastest slew rate. Read the average supply current (right scope) for the three capacitances and compare with $I = C_L V_{DD} f$. The energy taken from the supply per cycle is $C_L V_{DD}^2$, independent of R_{on}. Where is it dissipated?

Table 3.2: average supply current at 1 MHz

	$C_L = 10$ pF	$C_L = 100$ pF	$C_L = 1$ nF
$C_L V_{DD} f$	33 µA	330 µA	3.3 mA
simulation	34	330	3300

Solution

Step 1: Charge per cycle: on every rising edge the supply delivers $Q = C_L V_{DD}$ to the load capacitance; on the falling edge this charge flows to GND, not back into the supply.

Step 2: Average current: $I = Q \cdot f = C_L V_{DD} f$, e.g. 1 nF $\cdot\, 3.3$ V $\cdot\, 1$ MHz $= 3.3$ mA; for 10 pF and 100 pF: 33 µA and 330 µA.

Step 3: Simulation: 34 µA, 330 µA, 3300 µA. The small excess at 10 pF is cross current during the gate edge (task 3.5).

Step 4: Energy: the supply delivers $E = QV_{DD} = C_L V_{DD}^2$ per cycle. At the end of the rising edge the capacitor stores $\frac{1}{2} C_L V_{DD}^2$, so the other half has been converted to heat in the PMOS. This holds for

any resistance: a smaller R_{on} means a larger current for a shorter time.

Step 5: On the falling edge the stored $\frac{1}{2}C_L V_{DD}^2$ is converted to heat in the NMOS.

Answer: The energy $C_L V_{DD}^2$ per cycle is dissipated half in the PMOS (charging) and half in the NMOS (discharging). $P = C_L V_{DD}^2 f$ is the dynamic power of all CMOS logic.

Task 3.5 [S/D] Set $C_L = 1$ pF and reduce the predriver slew rate to $0.02\,\mathrm{V/ns}$. Observe the supply current. Both transistors conduct while $V_T < V_G < V_{DD} - V_T$ (0.6 V to 2.7 V), at $0.02\,\mathrm{V/ns}$ that is 105 ns per edge. Measure the average supply current and compare it with $C_L V_{DD} f = 3.3$ µA (simulation: 376 µA). Why do microcontroller input pins have Schmitt triggers, and why does a slowly changing analogue voltage on a digital input increase the current consumption?

Solution

Step 1: Conduction window: the NMOS conducts for $V_G > V_T = 0.6$ V, the PMOS for $V_G < V_{DD} - V_T = 2.7$ V. Between 0.6 V and 2.7 V both are on.

Step 2: Duration: $2.1\,\mathrm{V}/0.02\,\mathrm{V/ns} = 105$ ns per edge, i.e. 210 ns per period of 1 µs.

Step 3: During this time a cross-conduction (shoot-through) current flows directly from V_{DD} through both transistors to GND. In the simulation its peak is 5.8 mA (at $V_G \approx V_{DD}/2$).

Step 4: Average supply current: with the fastest slew rate about 6 µA, with $0.02\,\mathrm{V/ns}$ 376 µA, more than 100 times $C_L V_{DD} f = 3.3$ µA. The cross current, not the load, determines the power.

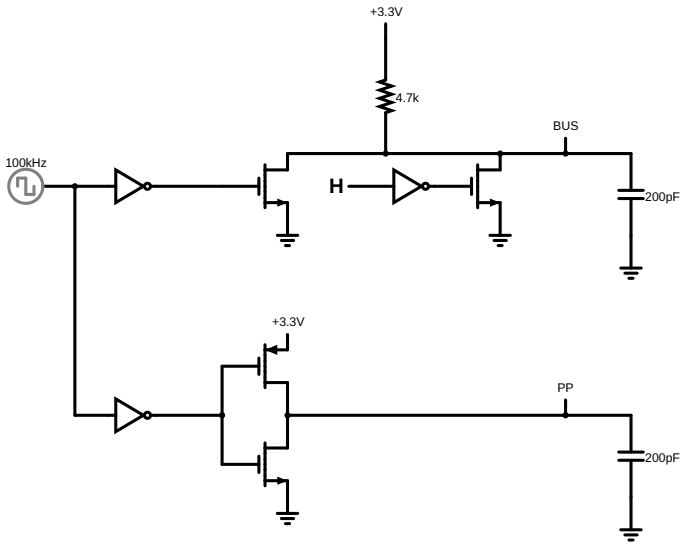
Step 5: Transfer to inputs: the input buffer of a pin is also a CMOS inverter. A slow input edge or an analogue voltage near $V_{DD}/2$ keeps both of its transistors on, permanently for a constant voltage.

Answer: A Schmitt trigger switches quickly and only once even for a slow or noisy input edge (hysteresis), so the cross current flows only briefly and the output does not oscillate. An analogue voltage on a digital input keeps the input buffer in its conduction window and causes a static cross current; unused or analogue pins should therefore have the digital input buffer disabled.

4 Open-drain bus versus push-pull

▶ open in Falstad (`D_open_drain_bus.txt`)

Two open-drain devices share a line with one pull-up resistor R_P, as on an I²C bus. Device 1 sends a clock; device 2 can hold the line low. A push-pull stage drives the same capacitance for comparison.



Task 4.1 [P] The I²C rise time is measured from 30 % to 70 % of V_{DD}. For the RC charge $V(t) = V_{DD}(1 - e^{-t/\tau})$ it is $t_r = \tau \ln(0.7/0.3) = 0.8473\, R_P C_b$, i.e. 796 ns for $R_P = 4.7\,\text{k}\Omega$ and $C_b = 200\,\text{pF}$.

Solution

Step 1: When the open-drain transistor switches off, only R_P charges the bus capacitance: $V(t) = V_{DD}(1 - e^{-t/\tau})$ with $\tau = R_P C_b$.

Step 2: Time to 30 %: $1 - e^{-t_{30}/\tau} = 0.3 \Rightarrow t_{30} = \tau \ln(1/0.7) = 0.357\,\tau$.

Step 3: Time to 70 %: $1 - e^{-t_{70}/\tau} = 0.7 \Rightarrow t_{70} = \tau \ln(1/0.3) = 1.204\,\tau$.

Step 4: Rise time: $t_r = t_{70} - t_{30} = \tau \ln(0.7/0.3) = 0.8473\,\tau$.

Step 5: Numbers: $\tau = 4.7\,\text{k}\Omega \cdot 200\,\text{pF} = 940\,\text{ns}$, $t_r = 0.8473 \cdot 940\,\text{ns} = 796\,\text{ns}$.

Answer: $t_r = 0.8473\, R_P C_b = 796$ ns, below the Standard-mode limit of 1000 ns, but far above the Fast-mode limit of 300 ns.

Task 4.2 [S] Measure the rise and fall times of BUS and the rise time of PP (same thresholds). Explain the difference between rising and falling edge of the open-drain line.

Table 4.1: edges with $R_P = 4.7\,\text{k}\Omega$, $C_b = 200\,\text{pF}$ (30 % to 70 %)

	t_r BUS	t_f BUS	t_r PP
model (RC)	$0.8473\, R_P C_b = 796$ ns	$0.8473\, R_{on,N} C_b \approx 5$ ns	$0.8473\, R_{on,P} C_b \approx 8$ ns
simulation	796	8	12

Solution

Step 1: Measure with the thresholds 0.99 V and 2.31 V ($30\,\%$ to $70\,\%$ of 3.3 V): BUS $t_r = 796$ ns, BUS $t_f = 8$ ns, PP $t_r = 12$ ns.

Step 2: The BUS rising edge matches the RC model exactly, because the pull-up is a real resistor.

Step 3: The falling edge of BUS and the rising edge of PP are somewhat slower than the RC estimate (5 ns, 8 ns), because the transistors start in saturation (task 3.1).

Step 4: Falling edge of BUS: the NMOS actively discharges C_b with about $31\,\Omega$. Rising edge: no transistor drives high, only $R_P = 4.7\,\text{k}\Omega$ charges C_b.

Step 5: Ratio: $4.7\,\text{k}\Omega / 31\,\Omega \approx 150$; the edges differ by a factor of about 100 (796 versus 8 ns).

Answer: The rising edge of an open-drain line is about 100 times slower than the falling edge, because it is driven only by the pull-up resistor. This asymmetry is the price for the wired-AND function.

Task 4.3 [P] Table 4.2 gives the permitted range of R_P for $C_b = 200$ pF and $V_{DD} = 3.3$ V according to the I²C specification (NXP UM10204): $V_{OL} \leq 0.4$ V at a sink current of 3 mA, $t_r \leq 1000$ ns (Standard-mode, 100 kHz) and $t_r \leq 300$ ns (Fast-mode, 400 kHz). Which limit comes from which requirement?

Table 4.2: pull-up range for $C_b = 200$ pF

	$R_{P,min} = \frac{V_{DD} - V_{OL}}{I_{OL}}$	$R_{P,max} = \frac{t_r}{0.8473\,C_b}$	standard values
Standard-mode	$967\,\Omega$	$5.9\,\text{k}\Omega$	$1\,\text{k}\Omega$…$5.6\,\text{k}\Omega$
Fast-mode	$967\,\Omega$	$1.77\,\text{k}\Omega$	$1\,\text{k}\Omega$…$1.5\,\text{k}\Omega$

Solution

Step 1: Lower limit from the low level: while a device pulls the line low, the current through R_P is $(V_{DD} - V_{OL})/R_P$. A device must only be able to sink 3 mA at $V_{OL} \leq 0.4$ V, so $R_P \geq (3.3\,\text{V} - 0.4\,\text{V})/3\,\text{mA} = 967\,\Omega$. A smaller R_P would demand more current than a compliant device guarantees.

Step 2: Upper limit from the rise time: $t_r = 0.8473\,R_P C_b \leq t_{r,max} \Rightarrow R_P \leq t_{r,max}/(0.8473\,C_b)$.

Step 3: Standard-mode: $R_{P,max} = 1000\,\text{ns}/(0.8473 \cdot 200\,\text{pF}) = 5.9\,\text{k}\Omega$.

Step 4: Fast-mode: $R_{P,max} = 300\,\text{ns}/(0.8473 \cdot 200\,\text{pF}) = 1.77\,\text{k}\Omega$.

Step 5: Standard values (E12) inside the ranges: Standard-mode $1\,\text{k}\Omega$…$5.6\,\text{k}\Omega$; Fast-mode $1\,\text{k}\Omega$…$1.5\,\text{k}\Omega$ ($1.8\,\text{k}\Omega$ is just above the limit, see task 4.4).

Answer: $R_{P,min}$ comes from the V_{OL}/sink current requirement (static), $R_{P,max}$ from the rise time (dynamic). The minimum is the same for both modes, the maximum is more than three times smaller for Fast-mode.

Task 4.4 [S] Check the limits in the simulation: complete Table 4.3. Then increase C_b to 400 pF (the maximum allowed by I²C) with $R_P = 4.7\,\text{k}\Omega$ (model: $t_r = 1.59\,\mu\text{s}$): which mode still works?

Table 4.3: low level and rise time over R_P ($C_b = 200$ pF)

R_P	5.9 kΩ	4.7 kΩ	1.8 kΩ	1 kΩ	470 Ω	220 Ω
I_{low}, model	0.56 mA	0.70 mA	1.8 mA	3.2 mA	6.6 mA	13.1 mA
V_{OL}, model	17 mV	22 mV	56 mV	99 mV	204 mV	406 mV
V_{OL}, simulation	17	22	56	101	211	437
t_r, model	1000 ns	796 ns	305 ns	169 ns	80 ns	37 ns
t_r, simulation	1000	796	306	168	78	36

Solution

Step 1: Model values: $I_{low} = V_{DD}/(R_P + R_{on,N})$, e.g. 220 Ω: $3.3\,\text{V}/251\,\Omega = 13.1$ mA; $V_{OL} \approx I_{low} R_{on,N}$, e.g. $13.1\,\text{mA} \cdot 30.9\,\Omega = 406$ mV; $t_r = 0.8473\,R_P C_b$.

Step 2: Simulation, low level: up to 1 kΩ model and simulation agree; at 470 Ω and 220 Ω the simulation is higher (211 and 437 mV), because R_{on} grows with V_{DS} in the triode region (the linear approximation only holds for small V_{DS}).

Step 3: Simulation, rise time: $t_r = 1000$ ns at 5.9 kΩ (exactly the Standard-mode limit) and $t_r = 306$ ns at 1.8 kΩ, just above the Fast-mode limit; 1.5 kΩ gives margin.

Step 4: Lower limit: our strong pin (31 Ω) keeps V_{OL} below 0.4 V down to about 250 Ω. The specification only guarantees 3 mA at 0.4 V, so 967 Ω remains the limit for a weak but compliant device: the bus must work with every device, not only with ours.

Step 5: $C_b = 400$ pF, $R_P = 4.7$ kΩ: $t_r = 0.8473 \cdot 4.7\,\text{k}\Omega \cdot 400\,\text{pF} = 1.59$ µs. Neither mode is met; at 100 kHz the line does not even reach V_{DD} (only 3.0 V).

Step 6: Required R_P at 400 pF: Standard-mode $\leq 1000\,\text{ns}/(0.8473 \cdot 400\,\text{pF}) = 2.9$ kΩ (possible, e.g. 2.7 kΩ); Fast-mode $\leq 885\,\Omega < R_{P,min} = 967\,\Omega$ (not possible with 3 mA devices).

Answer: With 400 pF and 4.7 kΩ no mode works. Standard-mode works with $R_P \leq 2.9$ kΩ. Fast-mode only works if all devices meet the additional rating of 6 mA at $V_{OL} = 0.6$ V ($R_{P,min} = 450\,\Omega$, e.g. 820 Ω), with Fast-mode Plus devices (20 mA), or if the bus is split by a buffer.

Task 4.5 [S/D] Set the input of device 2 to L. What happens to the clock of device 1? Which I²C feature uses this behaviour? What would happen with two push-pull outputs (part 2)?

Solution

Step 1: Observation: as long as device 2 holds the line low, BUS stays low, although device 1 releases it (high) every half period. The clock disappears from the line.

Step 2: Explanation: the line is high only if all devices release it. It is the AND of all outputs ("wired AND"). Device 2 pulling low simply overrides the pull-up; only a small current V_{DD}/R_P flows.

Step 3: Detection: device 1 can find out by reading back the line. If it released the line but reads low, another device is holding it.

Step 4: I²C uses this for clock stretching (a slow slave holds SCL low until it is ready; the master waits), for arbitration between masters (a master that sends 1 but reads 0 has lost and stops), and for the acknowledge bit (the receiver pulls SDA low).

Step 5: Push-pull: if device 1 drove high and device 2 low, the result would be the contention of part 2: about 28 mA, an undefined level of 1.09 V and up to 62 mW in one transistor.

Answer: The clock of device 1 is suppressed (wired AND). I²C uses this for clock stretching, arbitration and ACK. With push-pull outputs the same situation would be a short circuit.

Task 4.6 [D] While the line is low, R_P dissipates $P = V_{DD}^2/R_P$ (2.3 mW at 4.7 kΩ, 11 mW at 1 kΩ). Since $t_r = 0.8473\,R_P C_b$, the product $P \cdot t_r = 0.8473\,V_{DD}^2 C_b$ does not depend on R_P. What follows for the trade-off between speed and power of an open-drain bus, compared to push-pull? Propose a circuit that speeds up the rising edge without reducing R_P, and try it in the simulation (e.g. a current source pull-up or a switched pull-up).

Solution

Step 1: Static power while low: the full supply is across R_P: $P = V_{DD}^2/R_P$, i.e. $(3.3\,\text{V})^2/4.7\,\text{k}\Omega = 2.3\,\text{mW}$ and $(3.3\,\text{V})^2/1\,\text{k}\Omega = 11\,\text{mW}$.

Step 2: Product: $P \cdot t_r = \dfrac{V_{DD}^2}{R_P} \cdot 0.8473\,R_P C_b = 0.8473\,V_{DD}^2 C_b$. R_P cancels.

Step 3: Consequence: for a given bus (C_b, V_{DD}) speed and power can only be traded against each other. Halving the rise time doubles the static power.

Step 4: Push-pull: no static current (only $C_L V_{DD}^2 f$), both edges are driven actively. It is faster and more efficient, but it allows no wired AND.

Step 5: Faster rising edge without a smaller R_P: (1) a constant-current pull-up charges C_b linearly, so $t_r = 0.4\,V_{DD} C_b / I$ for 30 % to 70 %, and the current while low is still only I; (2) a rise-time accelerator that switches on a strong pull-up only while the line is rising above a threshold (e.g. LTC4311); (3) an active push-pull pull-up for the rising edge, as used by I²C Hs-mode masters.

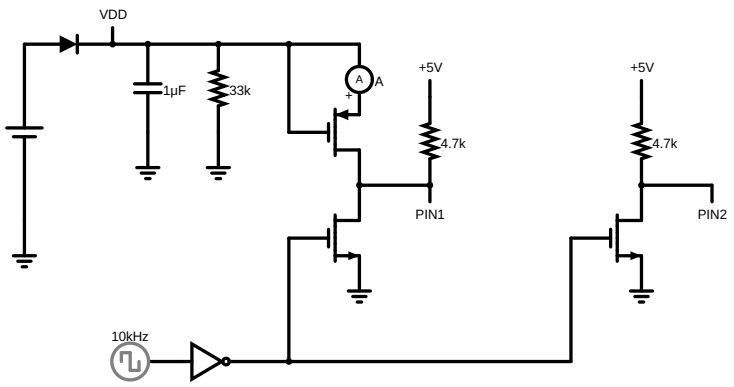
Step 6: In the simulation: replace R_P by a current source (e.g. 1 mA) and compare t_r and the current while low.

Answer: An open-drain bus with a resistor can only buy speed with static power. Active pull-ups (current source, accelerator, switched pull-up) break this trade-off, but they must never fight against a device that pulls low.

5 Maximum output voltage: open drain with a 5 V pull-up

► **open in Falstad** (`E_open_drain_overvoltage.txt`)

A 3.3 V microcontroller drives a 5 V logic input through an open-drain output and a pull-up to 5 V. PIN1 is a normal GPIO in open-drain mode: the PMOS is switched off, but it is still there. PIN2 is a 5 V-tolerant open-drain pin without a PMOS. The 3.3 V regulator can only source current (diode). The resistor "MCU load" represents the current consumption of the chip.



Task 5.1 [P/S] Table 5.1 gives the expected high levels and the current that flows from the 5 V supply into the 3.3 V supply via PIN1 ($R_{PU} = 4.7\,\text{k}\Omega$, diode forward voltage about 0.7 V). Explain the values, then measure them.

Table 5.1: open-drain pins with a pull-up to 5 V ($R_{PU} = 4.7\,\text{k}\Omega$)

	high level PIN1	high level PIN2	injection current
model	$V_{DD} + V_F \approx 4.0$ V	5.0 V	$\frac{5 - 4.0}{4.7}$ mA ≈ 0.2 mA
simulation	4.10	5.00	0.19

Solution

Step 1: PIN2 has no PMOS and therefore no diode to V_{DD}. When the NMOS is off, no current flows through the pull-up and the pin rises to the full 5.0 V.

Step 2: PIN1: the PMOS is switched off, but its body diode (pin $\rightarrow V_{DD}$) is still there. As soon as the pin rises above $V_{DD} + V_F$, the diode conducts and clamps the pin to $3.3\,\text{V} + 0.7\,\text{V} \approx 4.0$ V.

Step 3: Injection current: the rest of the 5 V drops across the pull-up: $I = (5\,\text{V} - 4.0\,\text{V})/4.7\,\text{k}\Omega \approx 0.2$ mA. It flows through the body diode into the 3.3 V supply.

Step 4: Simulation: PIN1 4.10 V, PIN2 5.00 V, injection current $(5 - 4.10)\,\text{V}/4.7\,\text{k}\Omega = 0.19$ mA. The clamp level is slightly higher than the estimate with 0.7 V.

Answer: The normal GPIO is clamped to about one diode drop above V_{DD} and feeds 0.2 mA into the 3.3 V rail; the 5 V-tolerant pin reaches 5 V without any current.

Task 5.2 [D] Is the high level of PIN1 sufficient for a 5 V CMOS input with $V_{IH} = 0.7 \cdot 5$ V? Is it allowed?

Solution

Step 1: Required high level: $V_{IH} = 0.7 \cdot 5\,\text{V} = 3.5$ V.

Step 2: PIN1 reaches $4.10\,\text{V} > 3.5$ V: logically sufficient, with 0.6 V margin.

Step 3: Absolute maximum rating: $IOVDD + 0.5\,\text{V} = 3.8$ V. PIN1 is 0.3 V above it, and a current flows through the body diode.

Step 4: The margin is not reliable: the diode voltage drops with temperature (about -2 mV/K) and with a smaller current (larger R_{PU}), and V_{DD} itself has a tolerance.

Answer: Logically yes, but it is not allowed: the pin is operated outside its absolute maximum rating, with a permanent injection current.

Task 5.3 [S/D] Set $R_{PU} = 1\,\text{k}\Omega$ and keep the MCU load at $33\,\text{k}\Omega$ ($100\,\mu$A, e.g. in a sleep mode). Observe V_{DD} (simulation: 3.9 V) and explain the effect (*back-powering*): the regulator can only source current, so V_{DD} rises until the chip consumes the injected current. What happens if the microcontroller is switched off ($V_{DD} = 0$) but the 5 V side stays on?

Solution

Step 1: Observation: V_{DD} rises to 3.9 V (with $470\,\Omega$: 4.0 V), above the absolute maximum supply voltage of the chip.

Step 2: Current balance at the 3.3 V rail: current flows in from the regulator (only in one direction, diode) and through the body diode of PIN1, and out through the MCU load.

Step 3: Injected current: while PIN1 is high, 0.25 mA flow through the body diode into the rail; since PIN1 is

high half of the time, the average is 0.12 mA.

Step 4: Equilibrium: the load draws only $V_{DD}/33\,\text{k}\Omega$. The regulator cannot sink the excess, so V_{DD} rises until the load consumes the injected current: $3.9\,\text{V}/33\,\text{k}\Omega = 0.12$ mA. The regulator then delivers nothing.

Step 5: Cross-check: with a load of 1 mA ($3.3\,\text{k}\Omega$) the chip consumes more than is injected; the regulator delivers the rest and the rail stays at 3.27 V. Back-powering is a problem especially in sleep modes.

Step 6: Microcontroller off ($V_{DD} = 0$): the chip is now powered only through its pin ("phantom powering"). It may start partially, hold the bus low, draw a large injection current, or come up in an undefined state when its supply returns.

Answer: The injected current back-powers the 3.3 V rail when the chip consumes less than is injected. A switched-off microcontroller is powered through its pin. Data sheets therefore forbid voltages above $V_{DD} + 0.3\,\text{V} \ldots 0.5\,\text{V}$ on non-tolerant pins, also when the chip is off.

Task 5.4 [D] Look up in the data sheets of an RP2040, an RP2350 and an STM32F4 which pins are 5 V tolerant and how large the injection current per pin and in total may be. Name two circuits that shift a 3.3 V open-drain signal to 5 V correctly, one of them bidirectional (for I^{2}C).

Solution

Step 1: RP2040: no GPIO is 5 V tolerant; the pin voltage is limited to $IOVDD + 0.5$ V (absolute maximum).

Step 2: RP2350: the GPIOs are 5 V tolerant while IOVDD is powered (and fail-safe when unpowered); the data sheet lists the exact conditions (e.g. not for pins used as ADC inputs).

Step 3: STM32F4: pins marked "FT" in the pin table are 5 V tolerant (in input or open-drain mode). The injection current is limited to about ± 5 mA per pin and ± 25 mA in total; positive injection is not allowed on FT pins.

Step 4: Level shifter 1 (unidirectional): a 5 V-tolerant pin in open-drain mode with a pull-up to 5 V, or an external open-drain transistor (part 6).

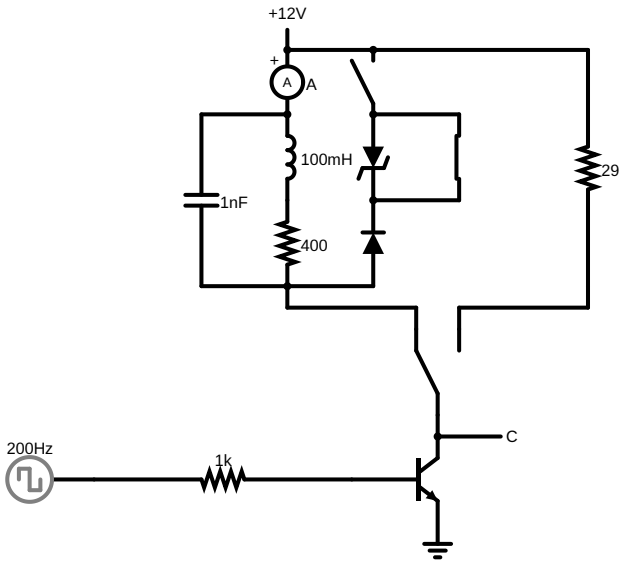
Step 5: Level shifter 2 (bidirectional, I^{2}C): one N-channel MOSFET (e.g. BSS138) per line, gate at 3.3 V, source on the 3.3 V side, drain on the 5 V side, pull-ups on both sides. If the 3.3 V side pulls low, the MOSFET turns on and pulls the 5 V side low; if the 5 V side pulls low, the body diode and then the channel pull the 3.3 V side low. Alternatively dedicated translators (e.g. PCA9306, TXS0102).

Answer: RP2040: no 5 V tolerant pins; RP2350: tolerant while powered; STM32F4: FT pins. A BSS138 per line with pull-ups on both sides shifts I^{2}C correctly in both directions.

6 Open collector: high voltage, high current, inductive loads

▶ open in Falstad (`F_open_collector_inductive.txt`)

A 3.3 V GPIO switches an NPN transistor ($\beta = 100$) through the base resistor R_B. The collector is an open-collector output for a 12 V load: a relay coil (0.1 H, $400\,\Omega$, 1 nF winding capacitance) or a 12 V/5 W lamp. The switch *flyback* adds a free-wheeling diode, opening the switch *bypass* puts a 24 V Zener diode in series with it. The timescale is shortened (switching at 200 Hz) so that you can see several cycles.



Task 6.1 [P] Table 6.1 gives the operating point with $R_B = 1\,\text{k}\Omega$. Is the transistor saturated?

Table 6.1: operating point, relay coil, $R_B = 1\,\text{k}\Omega$

	$I_C = \frac{12\,\text{V}}{400\,\Omega}$	$I_B = \frac{3.3 - 0.7}{R_B}$	βI_B	overdrive $\frac{\beta I_B}{I_C}$	V_{CE}
model	30 mA	2.6 mA	260 mA	8.7	< 0.2 V
simulation	29.8	2.61			0.07

Solution

Step 1: Required collector current: if the transistor were an ideal switch, the coil would see the full 12 V: $I_C = 12\,\text{V}/400\,\Omega = 30$ mA.

Step 2: Base current: the GPIO delivers 3.3 V, the base-emitter diode takes 0.7 V: $I_B = (3.3 - 0.7)\,\text{V}/1\,\text{k}\Omega = 2.6$ mA (the GPIO can deliver that easily, part 1).

Step 3: Possible collector current in the active region: $\beta I_B = 100 \cdot 2.6\,\text{mA} = 260$ mA.

Step 4: Overdrive: $\beta I_B / I_C = 260/30 = 8.7 > 1$. The transistor could conduct far more than the load allows, so V_{CE} drops to the saturation voltage (simulation: 0.07 V).

Answer: Yes, the transistor is saturated with an overdrive factor of 8.7; the simulation shows $I_C = 29.8$ mA and $V_{CE} = 0.07$ V.

Task 6.2 [S/D] Without flyback diode: when the transistor switches off, the coil current continues and charges the winding capacitance. Energy conservation ($\frac{1}{2}LI^2 = \frac{1}{2}CV^2$) gives a peak of $12\,\text{V} + I\sqrt{L/C} = 12\,\text{V} + 300\,\text{V}$. Measure it (simulation: _____296 V) and compare with V_{CEO} of a BC547 (45 V) or of a ULN2003 driver (50 V). What does the simulation not show?

Solution

Step 1: Before switching off, the coil stores $E = \frac{1}{2}LI^2 = \frac{1}{2} \cdot 0.1\,\text{H} \cdot (30\,\text{mA})^2 = 45\,\mu\text{J}$.

Step 2: The coil current cannot change abruptly. With the transistor off its only path is the winding capacitance, which is charged until all energy has moved into it: $\frac{1}{2}LI^2 = \frac{1}{2}C\,\Delta V^2$.

Step 3: $\Delta V = I\sqrt{L/C} = 30\,\text{mA} \cdot \sqrt{0.1\,\text{H}/1\,\text{nF}} = 30\,\text{mA} \cdot 10\,\text{k}\Omega = 300\,\text{V}$ above the 12 V supply, i.e. about 312 V at the collector.

Step 4: Simulation: peak $V_C = 296\,\text{V}$ (a little energy is lost in the coil resistance), followed by a ringing that drives the collector to $-14\,\text{V}$, where the base-collector diode conducts.

Step 5: Comparison: $V_{CEO} = 45\,\text{V}$ (BC547) or 50 V (ULN2003) is exceeded six times.

Step 6: What the simulation does not show: the transistor model has no breakdown. A real transistor breaks down (avalanche) at about 45 V to 80 V and absorbs the coil energy in a small volume, which degrades or destroys it. Neither the heating nor the EMI of the fast edge, nor the coupling back into the GPIO via the base, is modelled; the real winding capacitance and damping determine the actual peak.

Answer: Without a flyback diode the collector voltage reaches about 300 V, far above V_{CEO}. In reality the transistor would break down and be damaged; the simulation only shows the voltage the circuit would generate.

Task 6.3 [S] Close *flyback* and measure the maximum collector voltage and the time until the coil current has fallen to 10 % (right-click the scope, *Properties*, to change the time base). Then open *bypass* (diode + 24 V Zener) and measure again. Explain why the Zener variant is faster. Why does that matter for a relay?

Table 6.2: switching off the relay coil ($\tau = L/R = 0.25\,\text{ms}$)

	diode		diode + 24 V Zener	
	$V_{C,max}$	$t_{10\%}$	$V_{C,max}$	$t_{10\%}$
model	$12 + 0.7 = 12.7\,\text{V}$	475 µs	$12 + 24 + 0.7 = 36.7\,\text{V}$	87 µs
simulation	12.8	474	36.8	87

Model: $i(t) = -V_{cl}/R + (I_0 + V_{cl}/R)\,e^{-t/\tau}$ with the clamp voltage $V_{cl} = V_F$ or $V_Z + V_F$.

Solution

Step 1: Diode only, maximum voltage: the coil current continues through the diode to the 12 V rail, so the collector rises only one diode drop above it: $V_{C,max} = 12.7\,\text{V}$ (simulation 12.8 V).

Step 2: Decay: in the loop coil–diode, $L\,di/dt = -Ri - V_{cl}$ with $V_{cl} = V_F$. The solution is $i(t) = -V_{cl}/R + (I_0 + V_{cl}/R)\,e^{-t/\tau}$ with $\tau = L/R = 0.1\,\text{H}/400\,\Omega = 0.25\,\text{ms}$.

Step 3: Time to 10 % (3 mA) with $V_{cl}/R = 0.7\,\text{V}/400\,\Omega = 1.75\,\text{mA}$: $3\,\text{mA} = -1.75\,\text{mA} + 31.75\,\text{mA}\,e^{-t/\tau} \Rightarrow t = \tau\ln(31.75/4.75) = 475\,\mu\text{s}$ (simulation 474 µs; without V_F it would be $\tau \ln 10 = 576\,\mu\text{s}$).

Step 4: Diode + Zener, maximum voltage: the current now has to flow against $V_Z + V_F$: $V_{C,max} = 12 + 24 + 0.7 = 36.7\,\text{V}$ (simulation 36.8 V), still below $V_{CEO} = 45\,\text{V}$.

Step 5: Decay with $V_{cl} = 24.7\,\text{V}$, $V_{cl}/R = 61.8\,\text{mA}$: $i(t) = -61.8\,\text{mA} + 91.8\,\text{mA}\,e^{-t/\tau} = 3\,\text{mA} \Rightarrow t = \tau \ln(91.8/64.8) = 87\,\mu\text{s}$ (simulation $87\,\mu\text{s}$).

Step 6: Why faster: the energy $\frac{1}{2}LI^2$ is removed with the power $P = V_{cl} \cdot i$. With $24.7\,\text{V}$ instead of $0.7\,\text{V}$ across the clamp, the coil voltage $L\,di/dt$ is much larger, so the current falls much faster.

Answer: The Zener variant releases the relay about five times faster ($87\,\mu\text{s}$ instead of $475\,\mu\text{s}$) at the cost of a higher collector voltage. For a relay the current decay determines how fast the contacts open: a slow decay lets the contacts drift apart slowly, which increases arcing and contact wear. The designer trades transistor voltage rating against release speed.

Task 6.4 [S/D] Increase R_B to $47\,\text{k}\Omega$: $I_B = 55\,\mu\text{A}$, so $I_C \le \beta I_B = 5.5\,\text{mA}$ and $V_{CE} \approx 12 - 5.5 \cdot 0.4 = 9.8\,\text{V}$. Measure I_C and V_{CE} (simulation: _____5.7 mA and _____9.7 V). Will the relay still pull in?

Solution

Step 1: Base current: $I_B = (3.3 - 0.7)\,\text{V}/47\,\text{k}\Omega = 55\,\mu\text{A}$.

Step 2: Maximum collector current: $\beta I_B = 5.5\,\text{mA} < 30\,\text{mA}$ required: overdrive $0.18 < 1$, so the transistor is not saturated but in the active region (current source).

Step 3: Collector voltage: the coil drops $5.5\,\text{mA} \cdot 400\,\Omega = 2.2\,\text{V}$, so $V_{CE} = 12\,\text{V} - 2.2\,\text{V} = 9.8\,\text{V}$.

Step 4: Simulation: $I_C = 5.7\,\text{mA}$, $V_{CE} = 9.7\,\text{V}$. The transistor dissipates $9.7\,\text{V} \cdot 5.7\,\text{mA} = 55\,\text{mW}$ instead of about $2\,\text{mW}$ when saturated.

Step 5: The coil receives $5.7/30 = 19\,\%$ of its nominal current; a relay typically needs $70\,\%$ to $75\,\%$ to pull in.

Answer: No, the relay will not pull in. Rule: choose R_B so that $\beta_{min} I_B \ge 2 \ldots 5 \cdot I_C$ (overdrive), using the minimum β of the data sheet, not the typical one.

Task 6.5 [S/D] Switch the load to the lamp ($12\,\text{V}/5\,\text{W}$, needs $414\,\text{mA}$) and complete Table 6.3. A small-signal transistor in SOT-23 tolerates about $250\,\text{mW}$. Why is $330\,\Omega$ still not a good solution? Propose a better driver.

Table 6.3: lamp load

	I_B	I_C	V_{CE}	$P = V_{CE} I_C$
$R_B = 1\,\text{k}\Omega$, model	$2.6\,\text{mA}$	$\beta I_B = 260\,\text{mA}$	$12 - 0.26 \cdot 29 = 4.5\,\text{V}$	$1.2\,\text{W}$
$R_B = 1\,\text{k}\Omega$, simulation	2.56	256	4.57	1.17
$R_B = 330\,\Omega$, model	$7.9\,\text{mA}$	$414\,\text{mA}$ (saturated)	$\approx 0.1\,\text{V}$	$\approx 50\,\text{mW}$
$R_B = 330\,\Omega$, simulation	7.72	410	0.12	0.05

Solution

Step 1: Lamp: $I = P/V = 5\,\text{W}/12\,\text{V} \approx 414\,\text{mA}$, hot resistance $R = V^2/P = 28.8\,\Omega$.

Step 2: $R_B = 1\,\text{k}\Omega$: $I_B = 2.6\,\text{mA}$, $\beta I_B = 260\,\text{mA} < 414\,\text{mA}$: active region. The lamp drops $0.26\,\text{A} \cdot 28.8\,\Omega \approx 7.5\,\text{V}$, so $V_{CE} = 4.5\,\text{V}$ and $P = 4.5\,\text{V} \cdot 260\,\text{mA} = 1.2\,\text{W}$. Simulation: $256\,\text{mA}$, $4.57\,\text{V}$, $1.17\,\text{W}$. The lamp stays dim, and the transistor ($250\,\text{mW}$) overheats within seconds.

Step 3: $R_B = 330\,\Omega$: $I_B = 2.6\,\text{V}/330\,\Omega = 7.9\,\text{mA}$, $\beta I_B = 790\,\text{mA} > 414\,\text{mA}$: saturated, $V_{CE} \approx 0.1\,\text{V}$, $P \approx 0.12\,\text{V} \cdot 0.41\,\text{A} = 50\,\text{mW}$. Simulation: $7.72\,\text{mA}$, $410\,\text{mA}$, $0.12\,\text{V}$, $0.05\,\text{W}$.

Step 4: Why $330\,\Omega$ is still not good: (1) $7.7\,\text{mA}$ is a substantial load for the GPIO; V_{OH} drops (part 1) and the

per-domain budget shrinks. (2) The overdrive is only $790/414 = 1.9$ with the typical β; with the minimum β of the data sheet, and since β falls at high currents, the transistor may leave saturation. (3) A cold filament draws about ten times the nominal current when switched on.

Answer: A bipolar small-signal transistor is the wrong component for this load. Better: a logic-level N-MOSFET (e.g. with $R_{DS(on)} < 0.1\,\Omega$ at $V_{GS} = 2.5$ V, gate resistor about $100\,\Omega$, pull-down 100 kΩ), which needs no static gate current; a Darlington array such as the ULN2003 (500 mA, integrated clamp diodes); or a high-side switch with current limit.

Task 6.6 [D] Summarise: what limits the maximum output voltage and the maximum output current of a push-pull pin, of an open-drain pin and of an external open-collector stage?

Solution

Step 1: Push-pull pin: the output voltage cannot exceed V_{DD} (the PMOS connects to V_{DD}); an external voltage is clamped at $V_{DD} + V_F$ by the body diodes (part 1, part 5). The current is limited by R_{on} (level drop), by I_{sat} (short circuit), and by the per-pin and per-domain ratings (heat, electromigration).

Step 2: Open-drain pin: the high voltage is set by the pull-up, but limited by the pin structure: body/ESD diode to V_{DD} on a normal pin, about 5.5 V on a 5 V-tolerant pin. The sink current is limited by the NMOS as for push-pull; the high level is only as strong as the pull-up (slow edges, part 4).

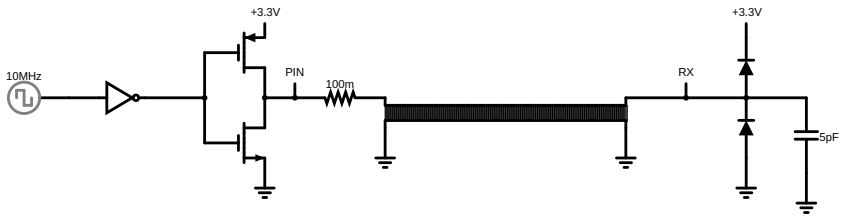
Step 3: External open collector/drain: the voltage is limited by V_{CEO} or $V_{DS,max}$ of the transistor, including inductive kick-back (part 6). The current is limited by βI_B (or V_{GS} and $R_{DS(on)}$ for a MOSFET), the saturation voltage and the power dissipation of the package.

Answer: Push-pull: V_{DD} and the on-chip transistors; open drain: the pin structure and the NMOS; external stage: the ratings of the external transistor, which can be chosen for the load.

7 Fast edges on a PCB trace

▶ open in Falstad (`G_transmission_line.txt`)

The push-pull pin drives a $50\,\Omega$ trace with 1 ns delay (about 15 cm on FR4). At the end there is the input of another chip: 5 pF and ESD diodes to V_{DD} and GND. R_S is a series resistor at the driver. The output stage is set to its highest drive strength and fastest slew rate, modelled by $\beta_N = 30$ mA/V^2 and $\beta_P = 25$ mA/V^2. The slew rate is adjustable as on many microcontrollers (RP2040: `SLEWFAST`, STM32: `OSPEEDR`).



Task 7.1 [P] A trace is "electrically long" if the edge is shorter than about twice its delay. Table 7.1 gives the values for $R_S = 0$ and the fastest slew rate. Explain the sequence: launched step, doubling at the open end, clamping, reflection at the driver.

Table 7.1: reflections for $R_S = 0$ ($R_{on,P} = 1/(25\,\text{mA/V}^2 \cdot 2.7\,\text{V}) = 15\,\Omega$)

edge time / round trip	< 1 ns / 2 ns: electrically long
$\Gamma_S = (R_{on} + R_S - Z_0)/(R_{on} + R_S + Z_0)$	$(15 - 50)/(15 + 50) = -0.54$
Γ_L (open end, 5 pF only rounds the edge)	$\approx +1$
launched step $V_{DD}\, Z_0/(Z_0 + R_{on})$	2.54 V
first voltage at RX (before clamping)	$2 \cdot 2.54\,\text{V} = 5.1\,\text{V}$
matched series resistor $R_S = Z_0 - R_{on}$	$\approx 36\,\Omega$ (33 Ω standard value)

Solution

Step 1: Electrically long: in its fastest setting the output edge takes well below 1 ns; the round trip (there and back) takes $2 \cdot 1\,\text{ns} = 2\,\text{ns}$. The edge is shorter than the round trip, so the line voltage is not the same everywhere: the trace must be treated as a transmission line.

Step 2: Source impedance: the strong PMOS has $R_{on,P} = 1/(25\,\text{mA/V}^2 \cdot 2.7\,\text{V}) = 15\,\Omega$ (NMOS: 12 Ω).

Step 3: Launched step: at the first moment the driver only sees Z_0, a voltage divider of R_{on} and Z_0: $3.3\,\text{V} \cdot 50/(50 + 15) = 2.54\,\text{V}$ travels down the line.

Step 4: Open end: the input (5 pF) is nearly open, $\Gamma_L \approx +1$. The wave is reflected with the same sign and the voltage at RX doubles to $2 \cdot 2.54\,\text{V} = 5.1\,\text{V}$.

Step 5: Clamping: 5.1 V is 1.8 V above V_{DD}, so the ESD diode to V_{DD} conducts and limits RX to about $V_{DD}+$ 0.6 V. The diode current flows into the supply of the receiver.

Step 6: Reflection at the driver: the returning wave sees $\Gamma_S = (15 - 50)/(15 + 50) = -0.54$. It is reflected with opposite sign and arrives at RX 2 ns later as an undershoot of roughly $-0.54 \cdot 2.5$ V, which is again limited by the ESD diode to GND.

Answer: The edge rings with a period of 4 ns (two round trips) and decays by the factor Γ_S per round trip, while the ESD diodes clamp the peaks.

Task 7.2 [S] Measure overshoot and undershoot at RX and the peak currents in the ESD diodes for the values of R_S in Table 7.2. Which R_S gives the cleanest edge? Compare with $R_S = Z_0 - R_{on}$.

Table 7.2: series termination (fastest slew rate)

R_S	0 Ω	10 Ω	22 Ω	36 Ω	50 Ω
$V_{RX,max}$, simulation	3.92	3.89	3.74	3.30	3.30
$V_{RX,min}$, simulation	-0.64	-0.61	-0.52	-0.02	0.00
I_{ESD} to V_{DD} / GND, simulation	30/40 mA	14/21	1/4	0	0

Solution

Step 1: $R_S = 0$: RX swings to 3.9 V and -0.64 V, beyond the ± 0.5 V limits. The ESD diodes conduct up to 30 mA (to V_{DD}) and 40 mA (to GND) on every edge, and the ringing can cross the input thresholds several times (double clocking).

Step 2: Measured values for all R_S:

R_S / Ω	0	10	22	36	50
$V_{RX,max}$ / V	3.92	3.89	3.74	3.30	3.30
$V_{RX,min}$ / V	-0.64	-0.61	-0.52	-0.02	0.00
I_{ESD} / mA	30/40	14/21	1/4	0	0

Step 3: With increasing R_S the launched step becomes smaller and Γ_S approaches zero, so overshoot and ESD currents decrease. At $36\,\Omega$ they vanish.

Step 4: Theory: with $R_S + R_{on} = Z_0$ the driver launches $V_{DD} \cdot Z_0/(2Z_0) = V_{DD}/2$; the open end doubles it to exactly V_{DD}; the reflected wave arrives at a matched source ($\Gamma_S = 0$) and is absorbed. $R_S = Z_0 - R_{on}$ lies between $50 - 15 = 35\,\Omega$ (rising) and $50 - 12 = 38\,\Omega$ (falling edge).

Step 5: $R_S = 50\,\Omega$ is also clean, but the source is now overdamped: the step at RX reaches V_{DD} only in several round trips, the edge becomes slower.

Answer: The cleanest edge is at $R_S \approx 36\,\Omega = Z_0 - R_{on}$ (standard value $33\,\Omega$), exactly as predicted by series termination. Larger values only slow the edge.

Task 7.3 [S/D] Set R_S back to 0 and reduce the slew rate. From which slew rate on does the ringing disappear? Why should you select the slowest slew rate that meets the timing of your interface?

Solution

Step 1: The slider sets the slew rate of the gate; the output switches during only part of the gate swing, so measure the edge at PIN, not at the gate.

Step 2: $1\,\mathrm{V/ns}$: the ringing is almost unchanged (ESD currents 19 mA/24 mA).

Step 3: $0.3\,\mathrm{V/ns}$: the ringing is reduced (4.5 mA/6.6 mA).

Step 4: $0.1\,\mathrm{V/ns}$: the output edge takes several ns, much longer than the 2 ns round trip. The reflections arrive while the edge is still rising and merge with it; the trace behaves like a lumped capacitance and the edge is clean.

Step 5: Criterion: the ringing disappears when the edge time is several times the round-trip time ($t_r \gtrsim 2 \ldots 3 \cdot 2t_d$).

Answer: The ringing disappears at about $0.1\,\mathrm{V/ns}$. Choose the slowest slew rate that meets the timing: a slow edge has a narrower spectrum, which means less ringing, less crosstalk, less radiated EMI, smaller ground bounce when many pins switch together, and less supply current (part 3).

8 Design task

Task 8.1 [P/S/D] A 3.3 V microcontroller board has to

(a) read a 5 V sensor over I^{2}C at 400 kHz; the cable is 30 cm long, the total bus capacitance is estimated at 250 pF,

(b) switch a 24 V/50 mA solenoid valve at up to 10 Hz,

(c) send a 10 MHz SPI clock over a 10 cm trace to a flash memory.

For each signal choose the output type and configuration of the pin (push-pull/open drain, drive strength, slew rate), dimension the external components and verify your choice in a simulation that you build from the given circuits. Document the remaining risks.

Solution

A possible solution (other justified solutions are equally valid):

(a) I^{2}C sensor at 5 V, 400 kHz, 250 pF:

Step 1: Output type: open drain is mandatory (wired AND, clock stretching, ACK).

Step 2: Level: the RP2040 pins are not 5 V tolerant $\Rightarrow$ a BSS138 level shifter per line (or a PCA9306) with pull-ups on both sides (task 5.4).

Step 3: Pull-up on the 5 V side, where the cable (250 pF) is: $R_{P,max} = 300\,\text{ns}/(0.8473 \cdot 250\,\text{pF}) = 1.42\,\text{k}\Omega$, but $R_{P,min} = (5\,\text{V} - 0.4\,\text{V})/3\,\text{mA} = 1.53\,\text{k}\Omega$: no resistor meets both.

Step 4: Options: devices rated for 6 mA at 0.6 V ($R_{P,min} = 4.4\,\text{V}/6\,\text{mA} = 733\,\Omega$, choose $1.2\,\text{k}\Omega$: $t_r = 254$ ns); a rise-time accelerator or bus buffer; or Standard-mode at 100 kHz ($R_{P,max} = 4.7\,\text{k}\Omega$, choose $1.8\,\text{k}\Omega$...$4.7\,\text{k}\Omega$). On the short 3.3 V side $4.7\,\text{k}\Omega$ is enough.

Step 5: Simulation: part 4 with a 5 V pull-up and 250 pF; check t_r and V_{OL}.

Step 6: Risk: the capacitance is only an estimate; measure t_r on the real board.

(b) Solenoid valve 24 V/50 mA at 10 Hz:

Step 1: Coil resistance $R = 24\,\text{V}/50\,\text{mA} = 480\,\Omega$. The GPIO cannot switch 24 V $\Rightarrow$ external low-side switch: logic-level N-MOSFET or a ULN2003 channel.

Step 2: Pin: push-pull, any drive strength, slow slew rate (10 Hz needs no fast edges); gate resistor $100\,\Omega$, pull-down $100\,\text{k}\Omega$ so that the valve stays off during reset (the pins are inputs then).

Step 3: Free-wheeling: diode + 24 V Zener if the valve must close quickly; then $V_{DS,max} \geq 24 + 24 + 1 \approx 50$ V, choose a 60 V MOSFET. Otherwise a diode alone ($V_{DS} \geq 40$ V with margin).

Step 4: Simulation: part 6 with 24 V, $480\,\Omega$ and the valve inductance.

Step 5: Risk: the valve inductance is unknown without a data sheet; measure the kick-back on the board.

(c) SPI clock 10 MHz over 10 cm:

Step 1: Output type: push-pull (open drain would be far too slow, part 4).

Step 2: Trace: 10 cm on FR4 has about 0.67 ns delay, 1.3 ns round trip. With the fastest edge it is electrically long (part 7).

Step 3: Edge budget: the half period is 50 ns, so an edge of a few ns is entirely sufficient.

Step 4: Configuration: moderate drive strength and slow slew rate, plus a series resistor $R_S = Z_0 - R_{on}$ (e.g. $33\,\Omega$) close to the MCU.

Step 5: Simulation: part 7 with the trace delay changed to 0.67 ns.

Step 6: Risk: ringing at the flash causes double clocking and corrupted data.